

Industrial 2-Finger Gripper – CAD Design & Stress Simulation

Mini Project / Assignment Problem Statement

1. Problem Statement

Design and analyze a **2-finger parallel jaw industrial gripper** suitable for robotic applications. The gripper must:

- Be fully modeled using parametric CAD tools
- Include fingers, base, and actuation interface
- Be capable of holding a payload between **5 kg and 10 kg**
- Pass static stress analysis under applied gripping forces

The final design must be validated using **static structural simulation**.

2. Tasks / Questions to Be Solved

Question 1: Conceptual Design

Develop a basic concept for a **parallel jaw gripper**.

Question: Propose a 2-finger gripper mechanism and explain how parallel motion of the jaws is achieved.

Deliverable: Hand sketch or digital concept diagram.

Question 2: CAD Sketching

Create **2D sketches** for all major components.

Question: Generate fully constrained sketches for the gripper base and fingers, showing key dimensions and constraints.

Question 3: 3D Modeling (Features)

Convert sketches into 3D parts using CAD features.

Question: Create 3D models of all gripper components using features such as extrude, cut, fillet, and chamfer.

Question 4: Assembly

Assemble all individual parts into a complete gripper.

Question: Assemble the gripper components and demonstrate correct alignment, motion constraints, and jaw symmetry.

Question: Calculate the gripping force required to safely hold a payload of 5–10 kg, considering friction and safety factor.

Question 5: Static Stress Simulation

Perform **static structural analysis** on critical components.

Question: Apply gripping forces and boundary conditions to the gripper model and evaluate stress, strain, and deformation.

3. General Guidelines

- The design must be **parametric**
- Reasonable assumptions for material properties must be stated
- Apply a **minimum safety factor of 1.5**
- Focus on clarity of modeling and correctness of simulation setup

4. Tools to Be Used

Preferred (If Available)

- **SolidWorks** (CAD + Simulation)
- **ANSYS Workbench** (Static Structural)

Free / Alternative Tools (Allowed)

If SolidWorks or ANSYS are **not available**, students may use:

- **Autodesk Fusion 360** (CAD + Simulation)
- **FreeCAD** (CAD)
- **SimScale** (cloud-based FEA)
- **Onshape** (CAD)

Use of free or open-source software is **fully acceptable** for this project.

5. Submission Guidelines

Files to Be Submitted

1. **Project Report (PDF)**
 - Design concept the color of gripper should be in yellow and blue color
 - CAD screenshots
 - Load calculations
 - Stress plots and deformation results
 - Final conclusion
2. **CAD Files**
 - Native CAD files (e.g., **.SLDPRT**, **.SLDASM**, **.f3d**, **.FCStd**)
 - OR neutral formats (**.STEP** / **.IGES**)
3. **Simulation Files / Results**
 - Screenshots of boundary conditions and results

Submission Format

- Single google drive named:

Industrial_Gripper_Design_<StudentName>.zip

Folder Structure

Industrial_Gripper_Design/
| — report.pdf
| — cad_files/
| — simulation_results/

6. Evaluation Criteria

- Quality of CAD design
- Correct assembly and constraints
- Accuracy of load calculations
- Correct stress simulation setup
- Interpretation of results
- Overall presentation and documentation